Arunabha Majumder

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Portfolio: arunabha22.github.io/arunabha-majumder

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Yuman, DTU Skylab, Lyngby

Application: Robotics Engineer

Dear Yuman team,

I am applying for the Robotics Engineer role. I am a mechatronics and robotics engineer doing my PhD at Aalborg University, and my research work is finished, so I can start soon while I complete my thesis. Robots that move small items around hospital wards appeal to me because they help nurses spend more time with patients, and because they have to work safely around people every day. That is exactly what I have designed for.

In my PhD I designed a hybrid shoulder exoskeleton where a variable stiffness mechanism and a small 6 Nm motor work together to support the arm. I optimised the design with a genetic algorithm, integrated the motor, encoder and force sensors, measured its stiffness on a physical test setup and tested it with 12 people. It cut energy use by more than 25% and won the Best Paper Award at IFToMM ISRM 2026. I also designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms, estimated its stiffness with less than 10% error and quantified human arm impedance. To run those tests, I implemented an Arduino-based trigger system to synchronise the robot's motion with Qualisys motion capture.

I am most useful where hardware meets the real world: building prototypes, integrating motors and sensors, finding out why something does not behave as planned, and fixing it. I have not worked on mobile robot navigation or ROS yet, and I would like to learn that side from your team while I help make the robots reliable for daily use in hospitals. I also hold a Senior Diploma in Fine Arts, so I explain ideas quickly with hand sketches.

I live in Denmark already and I am willing to learn Danish. Thank you for considering my application. You can see my work at arunabha22.github.io/arunabha-majumder.

Kind regards,
Arunabha Majumder